CanonFuse3D: Reliability-Gated Fusion of Uncalibrated Dual-View 3D Human Pose Streams for Fast-Moving Athletes

Abstract

Monocular three-dimensional (3D) human pose estimates remain vulnerable to depth ambiguity, self-occlusion, and temporal jitter. A second camera can provide complementary evidence, but exploiting it through calibrated reconstruction is difficult when cameras follow a distant, fast-moving athlete or are installed temporarily. We present CanonFuse3D, a calibration-free post-estimation framework that fuses two independently estimated, approximately synchronized monocular 3D pose streams without requiring images, 2D observations, or camera parameters at the fusion stage. Human-centric canonicalization and sequence-level alignment expose cross-view agreement and disagreement; a shared joint encoder, bidirectional cross-view attention, and temporal refinement then drive joint-frame gating with residual correction. We evaluate the framework quantitatively on a synthetic Unity dataset with avatar ground truth and the real-world Ski-PTZ-Pose dataset with a constructed reference, and qualitatively on author-collected dual-view recordings. Relative to canonical averaging, CanonFuse3D reduces the mean per-joint position error by 44.5% on Unity and 48.6% on Ski-PTZ-Pose, while comparisons with learned and geometric references reveal metric- and input-dependent trade-offs rather than universal superiority. Controlled studies characterize the useful range across camera separations, small timing and sampling-rate errors, and pose or image occlusion, including failures under severe corruption. The same architecture supports alternative pose estimators through lightweight adaptation and additional cameras through pairwise composition.

Keywords

3D human pose estimation; dual-view fusion; post-estimation pose fusion; calibration-free refinement; human-centric representation; front-end generalization